

UNIT 2 · Foundations of Maths, Linguistics & Knowledge Framework

2.1 Pāṇini & the Aṣṭādhyāyī (linguistics) **IMPORTANT**

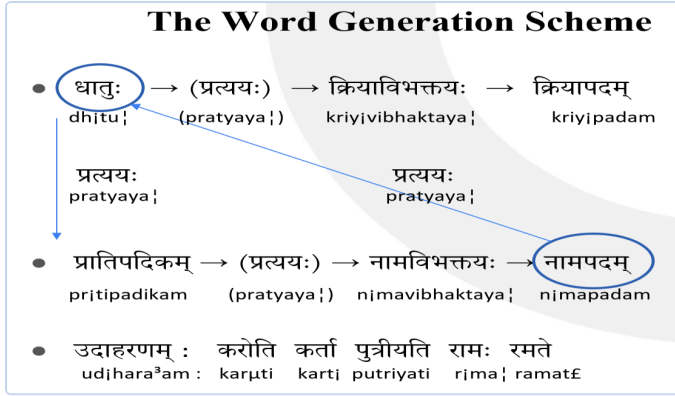
- **Pāṇini**: born in Gandhāra; c. 5th–6th c. BCE (some sources c. 400 BCE). **Father of linguistics**. His morphological analysis was ahead of any Western theory until the 20th century.
- **Aṣṭādhyāyī** = "eight chapters" (*aṣṭa* = 8, *adhyāya* = chapter): **3,983 sūtras**; 8 chapters × 4 quarters (*pādas*) = 32 sections. MQP Q14: Eight.
- Sūtra style = concise rules, memorised orally. Unique **metalanguage**; structure compared to **Backus–Naur Form (BNF)**.
- **Tri-Muni** of Sanskrit grammar: Pāṇini (*sūtras*) · Kātyāyana (*vārtikas*, 4th c. BCE) · Patañjali (*Mahābhāṣya*, 2nd c. BCE).
- Four parts of Pāṇinian grammar: **Śivasūtra/Maheśvara Sūtras** (14 phoneme sūtras; phonology; *pratyāhāra* shorthand) · Aṣṭādhyāyī (rules) · **Dhātupāṭha** (list of verbal roots, ~2,200) · **Gaṇapāṭha** (classes of noun stems).



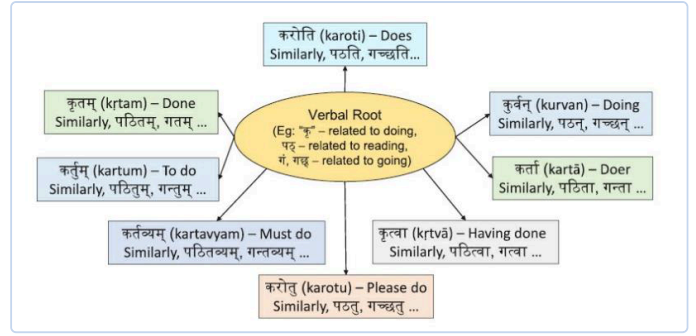
Pāṇini on the ₹500 stamp

Word generation: why it is "computational"

- **Root + suffix + rules → word**. Verbal root (*dhātu*) + suffix (*pratyaya*) → verb form; nominal stem (*prātipadika*) + **sup** suffix → noun form. Verb suffix = **tiṅ**. (SEE MCQs: noun declension suffixes = **sUP**; basic generator = **dhātu / prātipadika**.)
- Noun: 7 cases × 3 numbers = **21** sup suffixes. Verb: 3 persons × 3 numbers = **9** tiṅ suffixes.
- Example root *kṛ* (to do; in Dhātupāṭha "ḍukṛñ"): *karoti*, *kurvan*, *kartā*, *kṛtvā*, *karotu*, *kartavyam*, *kartum*, *kṛtam*. The **same pattern applies to ~2,200 verbs** (*paṭh* → *paṭhati*..., *gam* → *gacchati*...).
- Derivation is step-by-step and recursive: apply a rule if its conditions hold; repeat till no rule applies. **Vocabulary is not fixed: new valid words can be generated** → Sanskrit stays relevant; suited to NLP/AI.



Word generation scheme



Derivatives of root kr

Pāṇini's grammar	Computer science (CFG)
Sūtra (rule)	Production rule
Abstract categories (prātipadika, dhātu, sup, tiṅ)	Non-terminals (NP, VP)
Sounds / morphemes	Terminals
Sequential + recursive rule application (<i>anuvṛtti</i>)	Recursive expansion
Paribhāṣā sūtras (rules about rules)	Meta-grammar
Generative and transformative	Only generative

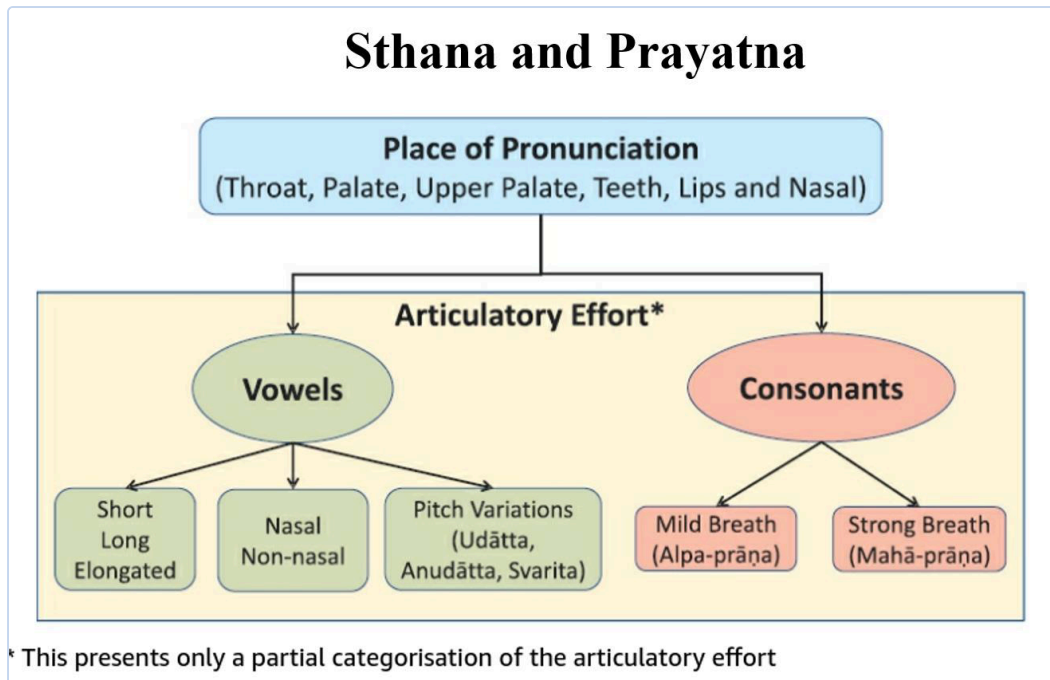
Example: *Rāmaḥ phalam khādati* = S → NP VP; NP → N SUFFIX; VP → V NP. Compound (samāsa): *tatpuruṣaḥ samāsaḥ, rāja-putra* = COMPOUND → N N.

IMPORTANT Why "father of linguistics"? Traditional: preserved Vedic Sanskrit accurately; 3,983 rule system; standard, universally accepted model. Modern: formal generative grammar; algorithmic, recursive, rule-based (BNF-like); influences computational linguistics, NLP, AI; oldest and most complete descriptive model of language.

Phonetics (Śikṣā / sound science)

- Linguistics = sound (listening/speaking) + script (reading/writing). Language-related Vedāṅgas: **Śikṣā** (phonetics), **Nirukta** (etymology), **Vyākaraṇa** (grammar).
- Sthāna** (MQP/SEE MCQ) = **place of articulation** (throat, palate, upper palate, teeth, lips, nasal); **Prayatna** = articulatory effort (e.g. mild breath/*alpaprāṇa* vs strong breath/*mahāprāṇa*; vowels short/long/nasal).

Sthana and Prayatna



Sthāna and Prayatna

[BOOK p.116 Pāṇini ↗](#)
[BOOK p.117 Phonetics ↗](#)
[BOOK p.120 Patterns ↗](#)
[BOOK p.122 Computational concepts ↗](#)
[BOOK p.128 Sentence logic ↗](#)

2.2 Number system, zero and decimal place-value IMPORTANT

- **Decimal place-value system** (10 symbols 0–9; position decides value) is India's greatest gift. Evidence: Ashokan edicts (3rd c. BCE); Brāhmī numerals; legal document **594 CE** and inscription **876 CE** (Gwalior) with place value and zero; Al-Bīrūnī's notes; Ifrah's research.
- Roman numerals are impractical: 432,000 needs M written 432 times.
- **Zero (śūnya)**: first formal definition word "śūnya" in Piṅgala's *Chandaḥśāstra*. **Brahmagupta (628 CE, Brahmasphuṭasiddhānta)**: zero as a **number** with rules ($a+0=a$, $a \times 0=0$; $a/0$ undefined; debts/fortunes). Āryabhaṭa used place value. Bakhshālī manuscript: dot for zero (carbon-dated 3rd–4th c. CE). Gwalior temple inscription (9th c.) = earliest recorded zero in stone.
- **Transmission**: India → Arabs (Kaṅka translated Sanskrit; **Al-Khwārizmī**, Al-Samaw'al) → Spain/Europe (Fibonacci's *Liber Abaci*, 1202). Words *algorithm*, *algebra*.
- **Large numbers**: Śukla Yajurveda 17.2 names powers of ten to 10^{12} (eka, daśa, śata, sahasra, ayuta, niyuta, prayuta, arbuda, nyarbuda, samudra..., parārdha). Pattern = *daśaguṇottara saṁjñā*. Lalitavistara: Buddha counts to *tallakṣaṇa* (10^{53}) and beyond (up to 10^{421}).
- **Measurement** (QB): three generic measures = **length, weight, time**. **Aṅgula** = basic Vedic unit (Śulba). Arthaśāstra: Dhanus = 96 aṅgula; Gṛhapatya-dhanus = 108 aṅgula (roads). Indus cubit $\approx$ 51.8 cm; Harappan brick ratio **4:2:1**; Harappan streets main 9–10 m, secondary 4–5 m, lanes 1.8–2.4 m.
- Mathematics at top of Vedāṅga Śāstras: "yathā śikhā mayūrāṇām... gaṇitam mūrdhani sthitam" = like the crest of a peacock (SEE MCQ).

Great mathematicians (SEE 5a: "any four")

Name · date	Work	Contribution
Śulba Sūtra authors (Baudhāyana, Āpastamba, Kātyāyana) · 800–600 BCE	Śulba Sūtras	Earliest geometry; Pythagorean theorem; $\sqrt{2}$; $\pi \approx 3.088$; altar constructions.
Piṅgala · c. 300 BCE	Chandaḥśāstra	Binary system, combinatorics, Pascal's triangle.
Āryabhaṭa · 476–550 CE	Āryabhaṭīya	Place value, square/cube roots, sine table, $\pi = 3.1416$, kuṭṭaka, Earth rotates.
Varāhamihira · 505–587 CE	Pañcasiddhāntikā, Bṛhat Saṃhitā	Sine table, $\sin^2 + \cos^2$, combinatorics, magic squares.
Bhāskara I · 7th c.	Laghu/Mahā-bhāskariya	Approx formula for sine; commentary on Āryabhaṭīya.
Brahmagupta · 598–668 CE	Brahmasphuṭasiddhānta	Zero and negative-number rules; algebra; quadratic; cyclic quadrilateral.
Mahāvīracārya · 800–870 CE	Gaṇita-sāra-saṅgraha	First dedicated maths textbook; permutations & combinations.
Bhāskara II · 1114–1185 CE	Līlāvati, Bijagaṇita, Siddhānta-śiromaṇi	Arithmetic & algebra; chakravāla; calculus ideas; $\pi = 3927/1250$.
Mādhava (Kerala) · c. 1350–1375	(Kerala school)	Infinite series for sin, cos, arctan, π (11 decimals), 250 years before Newton/Leibniz.

2.3 Unique ways to write numbers **IMPORTANT**

All three exploit poetry. **Rule:** "Aṅkānāṃ vāmato gatiḥ": **digits are read right to left** (the first word is the units digit).

(a) Bhūta-saṃkhyā: numbers as objects / beings / concepts

Bhūta-saṃkhyā system	
Number	Represented by (partial list only)*
0	Śūnya, Pūrṇa, Kha
1	Ādi, Candra, Prithivī, Eka
2	Aśvin, Pairs Of Limbs, Ayana, Dvandva, Dvi
3	Rāma, Guṇa, Loka, Kāla, Agni, Trinetra
4	Veda, Śruti, Yuga, Āśrama, Varṇa, Samudra, Kṛta
5	Bhūta, Śāstra, Bāṇa, Pāṇḍava, Indriya
6	Aṅga, Ṛtu, Darśana, Śanmukha, Ṣaṭ
7	Ṛṣi, Adri, Svara, Dhātu, Chandas
8	Vasu, Bhujāṅga, Siddhi, Dik, Kuñjara, Nāga
9	Gṛha, Aṅka, Nanda
10	Dik, Aṅgulī, Avatāra, Rāvaṇaśira
11	Rudra
12	Āditya, Rāśi
13	Viśva, Kāma
14	Manu
15	Tithi, Dina

Bhūta-saṃkhyā values (your notes)

Phrase	Digits	Number
veda-vedāṅka-candrāḥ	4,4,9,1	1944
khādri-rāmāgnayaḥ	0,7,3,3	3370
netra-kha-bhūmi-śaśi-sarpa	2,0,1,1,8	81102
bhujaṅga-nanda-dvi-naga-aṅga-bāṇa-ṣaṭ-kṛta-indavaḥ	8,9,2,7,6,5,6,4,1	146,567,298

CIE-1 values: manu 14, agni 3, elephant 8, tithi 15, rudra 11, Indra 3, Pāṇḍava 5. Quick set: eye/ pair 2; Rāma, agni, guṇa 3; Veda, yuga 4; bhūta, bāṇa, Pāṇḍava, indriya 5; ṛtu, aṅga 6; ṛṣi, adri 7; vasu, nāga, gaja, bhujaṅga 8; graha, nanda 9; kha/śūnya 0.

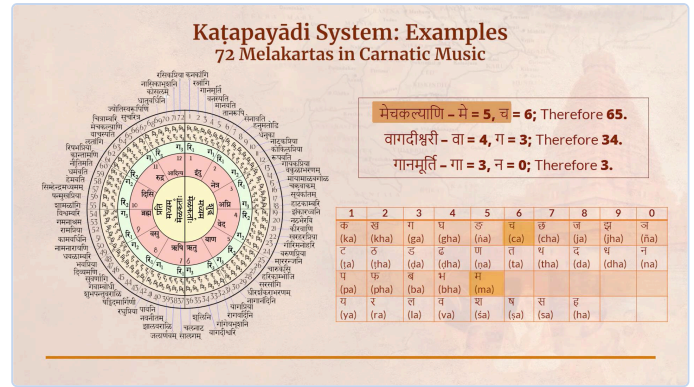
Trap: the Unit-2 QB gives khādri-rāmāgnayaḥ = 1733. By the rule in your notes (0,7,3,3 read right to left) it is **3370**. Write 3370 with working.

(b) Kaṭapayādi: consonants carry digits (mnemonic)

- Letters from **ka** start: ka-ña = 1...9,0; ṭa-na = 1...9,0; pa-ma = 1...5; ya-la = 1...9. **Standalone vowels = 0; in a conjunct only the last consonant counts; trailing bare consonant is ignored.** Read right to left.
- Examples: **bhavati** = bha 4, va 4, ti 6 → **644**. **śaktyāloke** = 5-1-3-1 → **1315**. **āyurārōgyasaukhyam** → 1,712,210.
- Famous: Śaṅkaravarman's *Sadratnamālā* verse "bhadrāmbudhisiddha..." encodes **π = 3.14159265358979324** (17 decimals). Also 72 *melakartā* rāgas: first two syllables give the rāga number (Mechakalyāṇī = 65).
- Advantage:** long numbers become easy-to-remember words/verses, ideal for oral tradition and codifying results.

1	2	3	4	5	6	7	8	9	0
क	ख	ग	घ	ङ	च	छ	ज	झ	ञ
ट	ठ	ड	ढ	ण	त	थ	द	ध	न
प	फ	ब	भ	म	-	-	-	-	-
य	र	ल	व	श	ष	स	ह	ळ	-

Kaṭapayādi letter-value table



72 melakartā example

(c) Āryabhaṭa system

- Consonants ka...ma = 1...25 (varga); ya...ha = 30, 40, 50, 60, 70, 80, 90, 100 (avarga). The 9 vowels give place values: a = 10^0 , i = 10^2 , u = 10^4 , ṛ = 10^6 , ṛ̣ = 10^8 , e = 10^{10} , ai = 10^{12} , o = 10^{14} , au = 10^{16} (steps of 100).
- Examples: **guṇa** = gu (3×10^4) + ṇa (15) = **30015**. **khyughṛ** = $(2+30) \times 10^4 + 4 \times 10^6 =$ **4,320,000** (a Mahāyuga). (Your notes print "2×30", which is a typo; it is 2+30.)

क/क	ख/ख	ग/ग	घ/घ	ङ/ङ	च/च	छ/छ	ज/ज	झ/झ	ञ/ञ
1	2	3	4	5	6	7	8	9	10
ट/ट	ठ/ठ	ड/ड	ढ/ढ	ण/ण	त/त	थ/थ	द/द	ध/ध	न/न
11	12	13	14	15	16	17	18	19	20
प/प	फ/फ	ब/ब	भ/भ	म/म					
21	22	23	24	25					
य/य	र/र	ल/ल	व/व	श/श	ष/ष	स/स	ह/ह		
30	40	50	60	70	80	90	100		
अ / आ	इ / ई	उ / ऊ	ऋ / ॠ	ॡ	ए	ओ	ऐ	औ	
10^0	10^2	10^4	10^6	10^8	10^{10}	10^{12}	10^{14}	10^{16}	

Āryabhaṭa letter values

Yuga cycle		
Yuga	Start (– End)	Length
<i>Krita (Satya)</i>	3,891,102 BCE	1,728,000 (4,800)
<i>Treta</i>	2,163,102 BCE	1,296,000 (3,600)
<i>Dvapara</i>	867,102 BCE	864,000 (2,400)
<i>Kali*</i>	3102 BCE – 428,899 CE	432,000 (1,200)
Years: 4,320,000 solar (12,000 divine)		

Yuga cycle (4,320,000 yrs)

Bhūta-saṃkhyā	Kaṭapayādi	Āryabhaṭa
Objects/concepts stand for digits	Consonants stand for digits; place value by position (right to left)	Each consonant = value; vowels = place value (powers of 100)
Uses whole words	Meaningful words from digits	Compact; used in Āryabhaṭīya

[BOOK p.146 Bhūta-saṃkhyā ↗](#)

[BOOK p.148 Kaṭapayādi ↗](#)

2.4 Piṅgala & the binary system **IMPORTANT**

- Chandaḥśāstra** by Piṅgala, c. **300 BCE** (textbook: 200–300 BCE; your Unit 2 notes: "2nd c. BCE"). Treatise on prosody (Chandas = Vedāṅga). Metre hierarchy: syllable → pāda → metre.
- Two syllable types: **Laghu (light/short, 1 mātrā)** and **Guru (heavy/long, 2 mātrās)**.
Guru has **4 varieties**: (1) long vowel; (2) short vowel before a conjunct consonant; (3) short vowel before anusvāra/visarga; (4) last syllable of a quarter (pāda).
- Convention: Laghu = 1, Guru = 0** (textbook Table 6.9 & 8.7, Unit 2 notes, SEE Feb 2026 MCQ "1 and 0").

Conflict: Unit 3 notes and CIE-2 solution say Laghu = 0 / Guru = 1. For MCQs use **Laghu = 1, Guru = 0** (SEE + textbook). In a written answer, state your convention explicitly.

Gaṇas: 3-syllable binary words, mnemonic **yamātārājabhānasalagām**

TABLE 6.9 Eight Gaṇas Defined by Piṅgala

Sl. No.	Gana Name	Binary Word*
1	'Ya' Gana	100
2	'Ma' Gana	000
3	'Ta' Gana	001
4	'Ra' Gana	010
5	'Ja' Gana	101
6	'Bha' Gana	011
7	'Na' Gana	111
8	'Sa' Gana	110

* 1 = Laghu; 0 = Guru

Textbook Table 6.9 (1 = Laghu, 0 = Guru)

Mnemonic for the 8 "gaṇas" of Piṅgala

S.No. Word	Gana Name	Binary
1	"Ya" Gana	100
2	"Ma" Gana	000
3	"Ta" Gana	001
4	"Ra" Gana	010
5	"Ja" Gana	101
6	"Bha" Gana	011
7	"Na" Gana	111
8	"Sa" Gana	110

यमाता-राज-भान-सलगम्
yamātā-rāja-bhāna-salagam



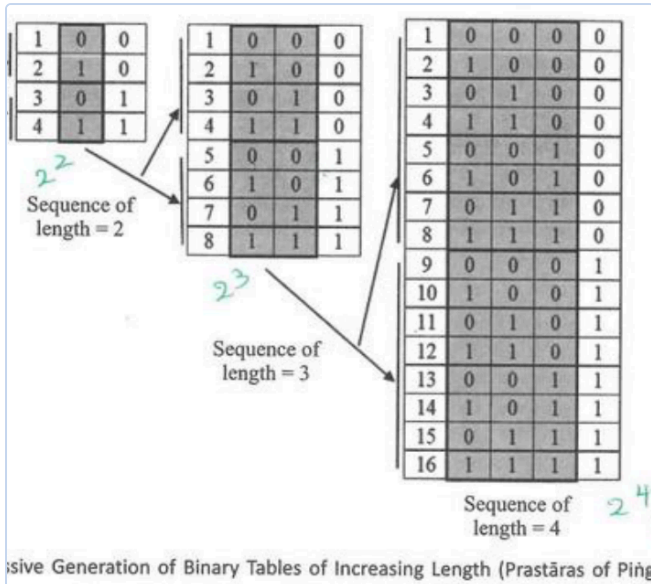
Mnemonic and binary cycle (your notes)

IMPORTANT Gaṇas and the de Bruijn sequence (SEE Feb 2026 4b)

- 8 gaṇas = all $2^3 = 8$ binary words of length 3: Ya 100, Ma 000, Ta 001, Ra 010, Ja 101, Bha 011, Na 111, Sa 110.
- The mnemonic *ya-mā-tā-rā-ja-bhā-na-sa-la-gam* converts to 1000101110; the last two digits "10" repeat the first two, so dropping them leaves a **cyclic binary cycle of length 8**: any 3 consecutive digits (cyclically) give a different gaṇa.
- This is exactly a **de Bruijn sequence B(2,3)**: every possible length-n string over a k-letter alphabet occurs exactly once as a substring ($k = 2$: laghu/guru; $n = 3$). Piṅgala did this ~2,300 years before de Bruijn (rediscovered recently). Applications: coding, robotics, cryptography.
- CIE-1: binary cycle of length 4 = 0000100110101111 (16 words).

Piṅgala's combinatorial toolkit

Term	Meaning
Prastāra	Listing all 2^n patterns as an array (built by doubling: copy the array, add a column with 0 in first half, 1 in second).
Samkhyā	Count of rows = 2^n .
Naṣṭa	Row number → pattern.
Uddiṣṭa	Pattern → row number.
Lagakriyā	How many patterns have r ones (nCr), via Varṇa-Meru = Pascal's triangle (rows 1; 1 1; 1 2 1; 1 3 3 1...), later rediscovered by Pascal (1655). Mātrā-meru gives Fibonacci-type counts.
Adhvayoga	Space the array occupies.



Prastāras of length 2, 3, 4 (textbook Fig 8.7)

[BOOK p.154 Piṅgala & binary ↗](#)

[BOOK p.156 Gaṇas & de Bruijn ↗](#)

[BOOK p.210 Chandaḥśāstra maths ↗](#)

[BOOK p.212 Naṣṭa/Uddiṣṭa ↗](#)

Worked algorithms (textbook p.212–213)

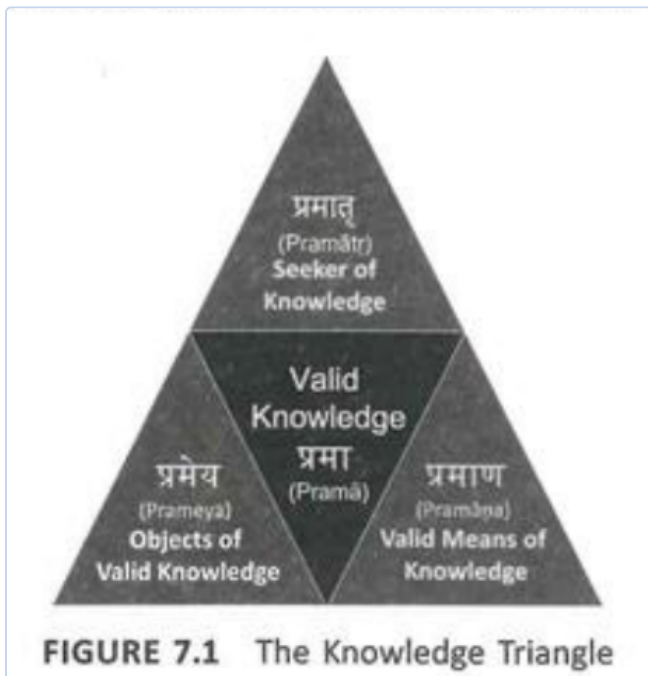
Naṣṭa: if the number is divisible by 2 write **1** and halve; if not, write **0**, add 1 and halve; repeat until length reached. Row 13 (length 4): $13 \rightarrow 0, 7 \rightarrow 0, 4 \rightarrow 1, 2 \rightarrow 1 \Rightarrow \mathbf{0011}$.

Uddiṣṭa: start at 1; scan from right; first '1' $\times 2$; for each next digit: '1' $\rightarrow$ double, '0' $\rightarrow$ double and subtract 1. $0111 \Rightarrow 15$.

Shortcut: row = (mirror of the binary word read as a number) + 1. 5th row of length 3 = $001 \rightarrow$ mirror $100 = 4 \rightarrow +1 = 5$.

2.5 Knowledge framework: Prameya, Pramāṇa, Saṃśaya IMPORTANT

Indian epistemology (Pramāṇa Śāstra): knowledge = true awareness acquired by a mode of inquiry. Two schools handle it: **Vaiśeṣika** (ontology: *what exists*, i.e. Prameya) and **Nyāya** (logic & epistemology: *how to know validly*, i.e. Pramāṇa).



The Knowledge Triangle (textbook Fig 7.1)

IMPORTANT Knowledge Triangle (MQP 4a)

- **Pramātr:** the seeker/knower.
- **Prameya:** object of valid knowledge (what is to be known).
- **Pramāṇa:** valid means of knowledge.
- Centre: **Pramā** = valid knowledge produced.
- **Saṃśaya** (doubt/ambiguity) is where inquiry **begins** (MQP Q18). Then Prayojana $\rightarrow$ argument & logic $\rightarrow$ Siddhānta (tenet) $\rightarrow$ Nirṇaya (conclusion).

Prameya (Vaiśeṣika: physical reality) · founder Kaṇāda, ~370 sūtras

- Method: **Uddeśa** (enumerate) $\rightarrow$ **Lakṣaṇa** (define) $\rightarrow$ **Parīkṣā** (examine by perception & inference).

- **Six padārthas (categories)**: **Dravya** (substance), **Guṇa** (attribute), **Karma** (action): objective/concrete; **Sāmānya** (universality/genus), **Viśeṣa** (specialty/species), **Samavāya** (inherence/combination): intellectual.
- **Nine dravyas**: Pṛthivī (earth), Ap (water), Tejas (fire), Vāyu (air), Ākāśa (ether), Dik (space), Kāla (time), Ātman (self), Manas (mind). 4 elements + mind finite and atomic; ākāśa, dik, kāla, ātman infinite.
- Atoms (*aṇu*): 2 aṇu = dvyaṇuka (dyad); 3 dyads = tryaṇuka (smallest visible).
- **Guṇa**: 17 listed by Kaṇāda, expanded to 24 (colour, taste, smell, touch, number, conjunction, cognition, pleasure...). **Karma** (5 types: throwing up, down, contraction, expansion, motion); action belongs to one substance, drives conjunction/disjunction.

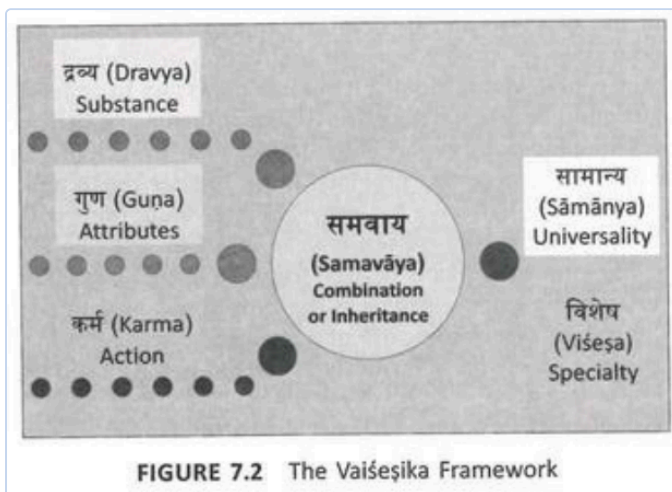


FIGURE 7.2 The Vaiśeṣika Framework

Vaiśeṣika framework (Fig 7.2)

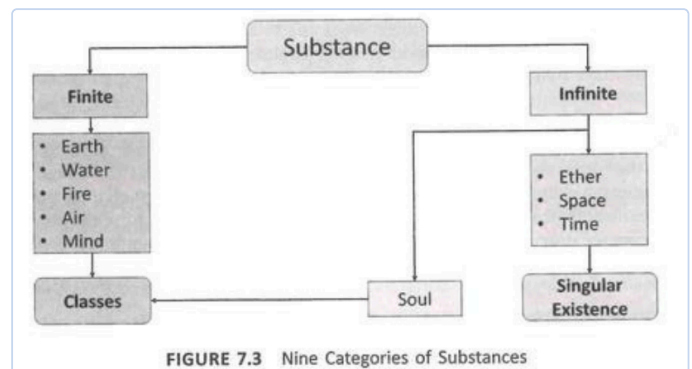


FIGURE 7.3 Nine Categories of Substances

Nine substances (Fig 7.3)

Pramāṇa: the four valid means (Nyāya) **IMPORTANT**

Pramāṇa	Meaning	Key points / example
Pratyakṣa (perception)	Direct knowledge via contact	5 senses + Manas (mind): sight, hearing, taste, smell, touch. Chain: Ātmā → Manas → Indriya → Artha. Most fundamental; all others rest on it. Types (CIE-1): laukika (ordinary) and alaukika (sāmānyalakṣaṇa, jñānalakṣaṇa, yogaja). Eg: counting 392 people.
Anumāna (inference)	Knowledge that "follows" perception	Smoke on hill → fire. Terms: Pakṣa (locus), Sādhya (what is to be proved), Hetu (reason), Vyāpti (invariable concomitance). 3 kinds: pūrvavat (a priori), śeṣavat (a posteriori), sāmānyatodṛṣṭa (commonly seen).
Upamāna (analogy)	Comparison / similarity	Gavaya looks like a cow → recognise a nilgai. Types: sādrśya (similarity), vaidharmya (dissimilarity), asādhāraṇa-dharma (special trait: rhinoceros horn).
Śabda (testimony)	Word of a reliable person (āpta)	<i>āptopadeśaḥ śabdaḥ</i> (Nyāya Sūtra 1.1.7). Āpta = right knowledge, willing to share, no intention to deceive, free from attachment/aversion. Eg: doctor's advice; Vedic injunctions.

IMPORTANT Pañca-avayava anumāna (5-step inference) (SEE Feb 2026 4a)

Example from your notes: "Force has been imparted to this ball because it is moving."

#	Step	Statement
1	Pratijñā (proposition)	The ball has the property of being subjected to force.
2	Hetu (reason)	Because it has motion.
3	Udāharaṇa (example)	Wherever there is motion there is force, like a cart.
4	Upanaya (application)	This ball is also so (has motion).
5	Nigamana (conclusion)	Hence the ball is subject to force.

Classic: Pratijñā "the hill has fire"; Hetu "because it has smoke"; Udāharaṇa "like a kitchen"; Upanaya "this hill has smoke"; Nigamana "so it has fire."

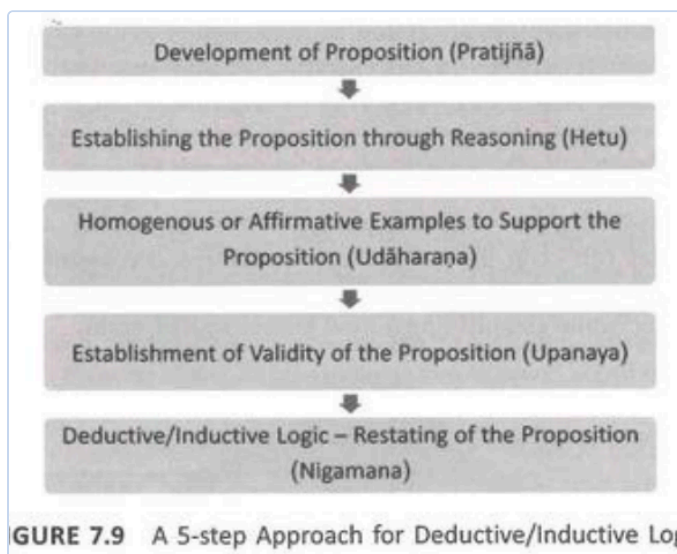
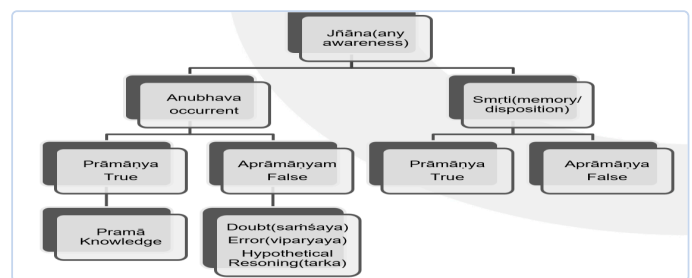


FIGURE 7.9 A 5-step Approach for Deductive/Inductive Logic

5-step logic (Fig 7.9)



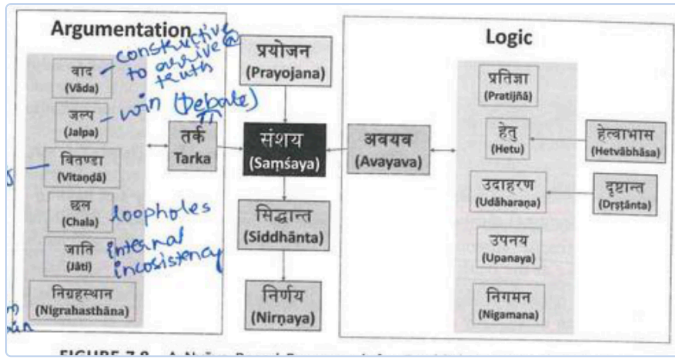
Jñāna tree: Anubhava/Smṛti → Pramā / Apramā (doubt, error)

Samśaya: ambiguity as the start of knowledge

- New knowledge begins by recognising ambiguity in existing knowledge. Five causes (textbook Fig 7.7): (1) several observed common properties; (2) uncommon/unknown properties with no explanation; (3) **conflicting findings**; (4) variation in results; (5) **non-apprehension** of the problem.

Framework for establishing valid knowledge (Nyāya)

- Tarka** = debate/reasoning in presence of ambiguity. Between a **vādin** (proponent) and **prativādin** (opponent). Six argument modes: **Vāda** (constructive, truth-seeking) · **Jalpa** (win the debate) · **Vitaṇḍā** (only attack, "I know you are wrong") · **Chala** (exploit loopholes) · **Jāti** (point to inconsistency) · **Nigrahasthāna** (raise objections / defeat points).
- Hetvābhāsa**: five fallacies: **Savyabhicāra** (erratic), **Viruddha** (contradictory), **Prakaraṇasama** (restates the problem; circular), **Sādhyaśama** (unproved), **Kālātīta** (mistimed / irrelevant).
- Siddhānta** = established tenet (4 kinds in Nyāya); **Nirṇaya** = final conclusion.



Nyāya framework (Fig 7.8; blue ink is from your book)



FIGURE 7.10 Potential Fallacies in Reasoning

Five fallacies (Fig 7.10)

- BOOK p.162 Tarka ↗
- BOOK p.163 Knowledge triangle ↗
- BOOK p.165 Prameya ↗
- BOOK p.167 Dravyas ↗
- BOOK p.170 Guṇa & Karma ↗
- BOOK p.173 Pramāṇa ↗
- BOOK p.176 Saṁśaya ↗
- BOOK p.178 Framework ↗
- BOOK p.181 Deductive logic ↗
- BOOK p.182 Fallacies ↗